

RESEARCH PORTFOLIO SERIES · CRITICAL ANALYSIS #04

Many Nuclei, Few Donors

Evaluating biological replication, multi-omic inference, and causal claims in a human ovarian-aging atlas

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Jin, C., Wang, X., Yang, J., et al. "Molecular and genetic insights into human ovarian aging from single-nuclei multi-omics analyses."

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01 Central Claim and Evidentiary Overview

A technically impressive atlas, built on a small number of people.

Jin et al. set out to build a molecular map of how the human ovary ages, using two sequencing technologies — single-nucleus RNA sequencing (snRNA-seq) and single-nucleus ATAC sequencing (snATAC-seq) — applied to the same eight frozen ovarian tissue samples. Four donors were young (23–29 years) and four were reproductively aged (49–54 years). From this tissue, the authors recovered 42,568 nuclei for the RNA side of the atlas and 41,550 nuclei for the chromatin-accessibility side. The paper's headline claims are that ovarian aging involves coordinated transcriptomic and epigenomic change across cell types, that mTOR signaling is a prominent and comparatively ovary-specific aging pathway, that a transcription factor called CEBPD becomes more active with age across most ovarian cell types, and that variants associated with age at natural menopause (ANM) can be mapped onto specific regulatory elements and candidate target genes in specific ovarian cell types.

My job in this analysis is not to summarize those claims — the abstract already does that efficiently — but to ask how much confidence each claim deserves, given the evidence actually presented. The central tension I want to sit with is this: the paper is genuinely large at the level of nuclei (tens of thousands of them) but genuinely small at the level of people (eight). Those two numbers describe different things, and treating them as if they carry the same evidentiary weight is, I think, the single most important interpretive trap this paper sets for a careless reader.

How much biological confidence can be placed in a technically detailed single-nucleus atlas when the molecular measurements include more than 84,000 nuclei but the age comparison is based on only eight human donors?

I want to be upfront about where I land, because I do not think the honest answer is "the paper is unreliable." Rare human ovarian tissue, collected under IRB-approved protocols from donors who died suddenly or underwent bilateral salpingo-oophorectomy, is difficult to obtain at scale. Eight donors with matched RNA and chromatin profiles is, in that context, a real accomplishment. What I want to examine carefully is which of the paper's claims are supported at the level that eight donors can support, and which claims quietly borrow strength from the large nucleus counts to sound more general than the donor-level evidence actually allows.

HOW I APPROACHED THIS ANALYSIS

I have not reanalyzed the raw single-nucleus data myself. My evaluation is based on the reported methods, results, extended data, figure legends, statistics and reproducibility statement, and data/code availability sections of the published article, read in full rather than only the abstract and discussion.

02 Theoretical Foundation

Human ovarian aging

The ovary is described in the introduction as the first human organ to show profound age-related decline, characterized by a progressive drop in follicle number and oocyte quality that accelerates around age 37. This decline is tied to fertility loss, aneuploidy risk, and, through age at natural menopause, to broader health outcomes including cardiovascular and bone health. What makes the ovary an unusual target for aging research is that its decline is comparatively fast and

visible relative to most other organs, which is part of why the authors frame it as a useful model system for studying aging biology more generally, not only reproductive biology.

snRNA-seq and snATAC-seq

Single-nucleus RNA sequencing captures messenger RNA present inside individual nuclei extracted from frozen tissue, which is useful for tissue types that are hard to dissociate into healthy live single cells. Single-nucleus ATAC sequencing instead measures which regions of chromatin are accessible — that is, physically open and therefore potentially available for transcription factors to bind — again nucleus by nucleus. Applied together, these two assays can, in principle, connect "which genes are expressed differently with age" to "which regulatory regions are more or less open with age," which is a genuinely powerful combination for generating mechanistic hypotheses. It is worth being precise, though, about what "together" means here: the paper performed these as two separate library preparations from the same frozen tissue samples, not as a single assay that reads both RNA and chromatin state out of the same physical nucleus. I return to why that distinction matters in Section 3.

Differential expression and chromatin accessibility

The paper identifies differentially expressed genes (DEGs) between young and aged ovaries within each of the eight snRNA-seq cell types, and differentially accessible regions (DARs) within each of the seven snATAC-seq cell types. Both used cell-level statistical testing (the MAST test, appropriate for the zero-inflated nature of single-cell/nucleus count data) with Bonferroni correction and a fold-change threshold. I flag this now because "cell-level testing" is a phrase worth remembering going into Section 4: the Methods describe nucleus-level testing but do not explicitly report donor-level aggregation or donor adjustment. This leaves uncertainty about how within-donor dependence was handled, which I examine in Section 4.

Regulatory networks and GWAS integration

The last conceptual layer of the paper links chromatin accessibility to gene regulatory networks (which transcription factor motifs sit in accessible regions near which genes) and then overlays this ovarian regulatory map onto genome-wide association study (GWAS) variants linked to age at natural menopause. The logic is that most GWAS-associated variants for complex traits like ANM sit in non-coding DNA, and that an ovary-specific regulatory map should help explain which of those variants might actually do something biologically, and in which ovarian cell type. This is a well-motivated and increasingly common strategy in the field, sometimes called "post-GWAS functional annotation," and the ovarian angle here is genuinely novel, since a cell-type-resolved epigenomic atlas of this tissue did not previously exist.

03 Data and Study Design

The study used fresh-frozen human ovarian tissue purchased from two tissue-procurement organizations, BioIVT and Cureline. Donors were either individuals who died suddenly or individuals who underwent bilateral salpingo-oophorectomy, with informed consent obtained from participants or their next of kin under IRB-approved protocols. Four donors were 23–29 years old ("young") and four were 49–54 years old ("reproductively aged," which the authors are careful to define as peri-menopausal to menopausal rather than simply "old"). All eight samples showed normal ovarian histology on hematoxylin and eosin staining, and post-mortem interval (PMI) related gene expression was assessed and found to have minimal effect on the transcriptome.

From this tissue, nuclei were isolated and processed on two parallel 10x Genomics workflows: one for snRNA-seq (3' gene expression kits) and one for snATAC-seq (ATAC-specific kits), each sequenced on an Illumina NextSeq 550. After quality-control filtering — nuclei with fewer than 200 genes, more than 6,000 genes, or more than 15% mitochondrial

reads were removed for the RNA side; nuclei outside defined ranges on peak-region fragments, reads-in-peaks percentage, blacklist ratio, TSS enrichment, and nucleosome signal were removed for the ATAC side — the study retained **42,568 nuclei for snRNA-seq** and **41,550 nuclei for snATAC-seq**. Unsupervised clustering identified eight cell types in the RNA data (stromal, blood endothelial, granulosa, smooth muscle, immune, lymphatic endothelial, epithelial, and theca cells) and seven in the ATAC data, where blood and lymphatic endothelial cells were combined into a single "endothelial cell" cluster and cell-type labels were transferred over from the RNA-based annotation.

I want to be careful here about a point the assignment brief specifically flagged, because it is easy to misread the paper's framing as claiming something stronger than it demonstrates. The RNA and chromatin measurements were not obtained as a simultaneous paired readout from the same individual nucleus. They are two separate assays performed on tissue from the same eight donors. That is still a real and useful design — it lets the authors compare cell-type composition and directionality of change between the two data types as an internal consistency check, and it does let them build cell-type-specific regulatory networks by overlaying accessible chromatin near a gene with that gene's expression change — but "matched tissue, separate assays" is a meaningfully different claim from "true single-cell multi-omic co-profiling," and the paper is, to its credit, accurate about this in its Methods, even where the abstract's phrase "single-nuclei multi-omics" could be read more ambitiously by someone skimming.

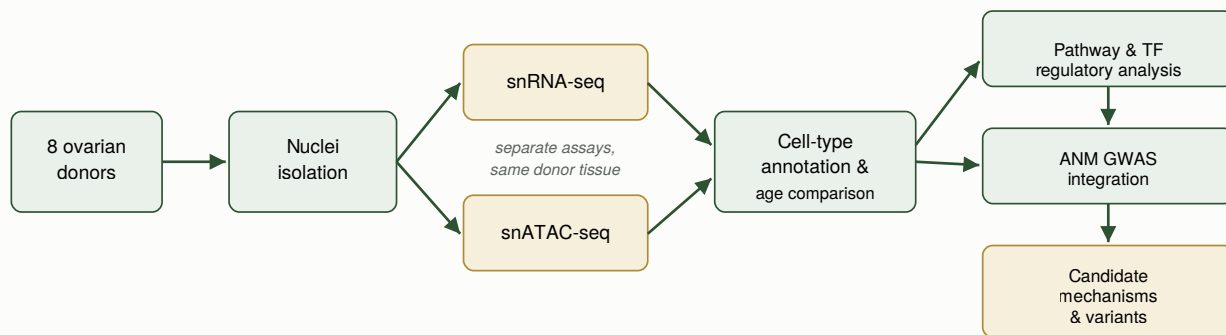


Figure 1. Study evidence pipeline (constructed by the author; not reproduced from Jin et al.). Eight ovarian donors were used to isolate nuclei, which were then processed through two separate assays — snRNA-seq and snATAC-seq — on the same donor tissue. Both were used to annotate cell types and identify age-associated changes, which then fed pathway/regulatory-network analysis and ANM GWAS integration, ultimately producing candidate mechanisms and variants.

04 From Thousands of Nuclei to Eight Donors — The Central Critique

Cellular depth versus biological replication

This is the crux of my analysis, so I want to state the hierarchy plainly. In this study, the biological replicates — the independent human beings whose ovarian aging is actually being measured — number eight, four per age group. The 42,568 and 41,550 nuclei are not additional biological replicates; they are additional *measurements*, nested within those eight people. A nucleus from Donor 3 is correlated with other nuclei from Donor 3 in ways a nucleus from Donor 7 is

not, because they share the same genotype, hormonal environment, medical history, and tissue-handling conditions. Having tens of thousands of nuclei genuinely does improve the resolution with which cell types can be identified and characterized — I do not want to undersell that — but it does not, by itself, tell you whether an aging-associated pattern would replicate in donor nine, ten, or fifty. That question can only be answered by adding more donors.

Cross-sectional age groups versus an aging trajectory

The design compares two non-overlapping age windows — 23–29 and 49–54 — captured at a single point in time from each donor. This lets the authors say something about how these two groups differ, on average, in the cell types and molecular signatures they present. It cannot, on its own, tell us when in the intervening two decades these changes begin, whether they accumulate gradually or shift abruptly around a particular life event (such as the perimenopausal transition), or how much individuals vary in their trajectory through that window. A longitudinal design — repeated ovarian sampling from the same individuals over time — would be the more direct way to resolve trajectory, though I recognize this is close to impossible in practice given how invasive ovarian tissue collection is; the study's reliance on tissue from deceased donors or women already undergoing oophorectomy for clinical reasons is itself a reasonable adaptation to that constraint, not a design flaw the authors could easily have avoided.

Age, reproductive status, and other possible confounding

Because the "aged" group spans the peri-menopausal to menopausal window, chronological age, reproductive/hormonal status, and follicle depletion are naturally bundled together in this design. A 51-year-old donor is simultaneously older, closer to or past menopause, and likely to have fewer follicles than a 26-year-old donor — and the study cannot cleanly separate which of these three, biologically entangled factors is driving a given molecular signature. The paper does not claim to resolve this, and I think that is the right call; but a reader skimming the abstract's language about "ovarian aging" could easily conflate the pathway changes reported here with pure chronological/cell-intrinsic aging, when part of what is being captured is plausibly the downstream consequence of a changed endocrine and follicular environment.

Statistical independence and possible pseudoreplication

I went through the Methods and figure legends specifically to see whether each analysis models donor identity appropriately or effectively treats nuclei as independent units. The picture is mixed, and I want to be fair about it rather than accuse the paper broadly of pseudoreplication, because at least one analysis is clearly donor-level. The cell-type proportion plots in Figure 2a-b report four biologically independent samples per age group, making the donor structure visible. However, the figure legend also describes a permutation test performed on cells, with 10,000 iterations. The reporting does not clearly establish whether permutations were constrained at the donor level, so I cannot confidently classify this test as fully donor-aware from the published description alone. The RNAscope in situ hybridization validation of RICTOR, MT-ATP6, and CDKN1A is on firmer ground: it is explicitly summarized at $n = 4$ biologically independent samples per group with a two-tailed unpaired t-test, which is unambiguously donor-level.

Where I am less certain is the differential expression and differential accessibility testing (MAST, used for DEGs, DARs, and differential motif activity) and the pathway-activity scoring (a moderated t-test with Benjamini-Hochberg correction). The MAST framework is designed for cell/nucleus-level count data. The Methods describe nucleus-level testing but do not explicitly report donor-level aggregation or donor adjustment — the published description does not state that donor identity was included as a random effect or covariate, nor does it describe a pseudobulk aggregation step (summing or averaging counts per donor before testing) for these comparisons. If nuclei rather than donors are the effective unit of statistical power in these tests, the very large sample sizes (thousands of nuclei per cell type) could make p-values look more significant than the underlying donor-level evidence would support — the classic pseudoreplication

concern. I want to be careful not to overstate this: I have not reanalyzed the data, so I cannot confirm this is what happened, only that the Methods as published do not clearly rule it out for these specific tests. A donor-aware sensitivity analysis — for example, a leave-one-donor-out check, or a mixed-effects model with donor as a random effect, or explicit pseudobulk differential expression alongside the cell-level test — would have let me assess this directly rather than infer it from what is and is not stated.

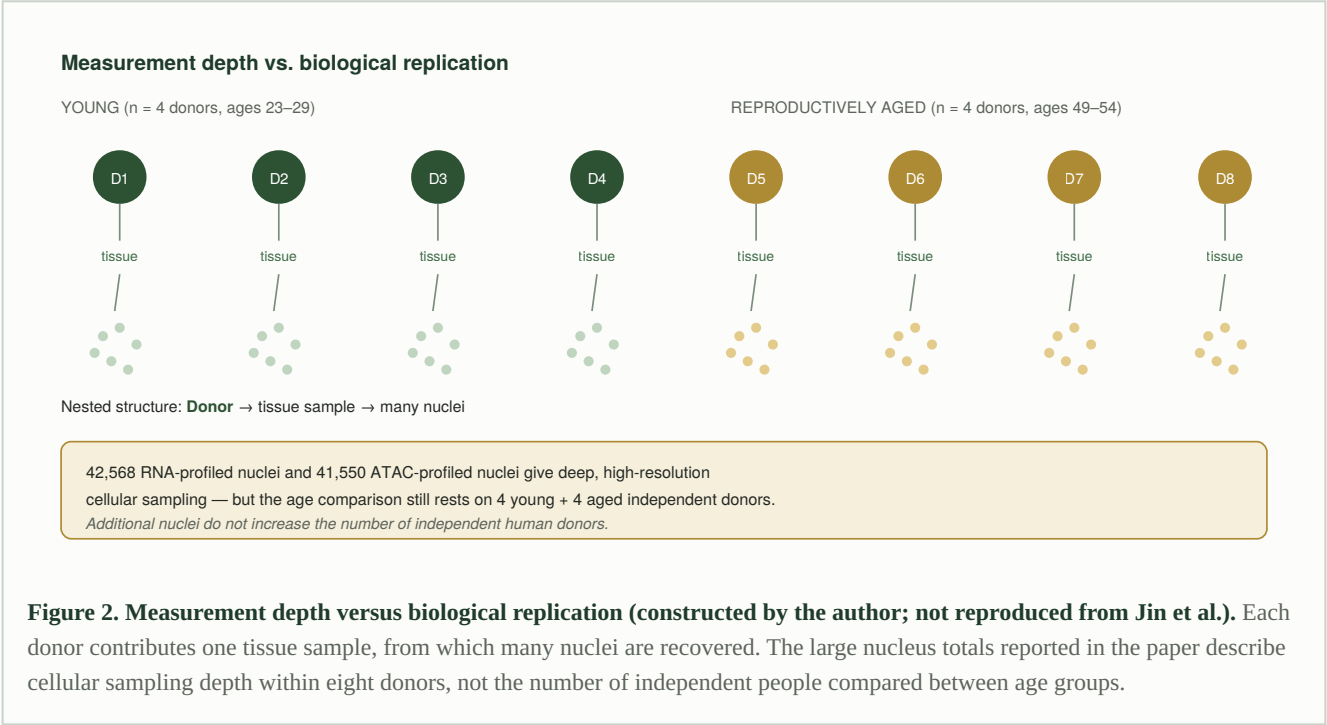


Table 3. Levels of replication in the ovarian aging comparison

Quantity	Young group	Aged group	Note
Number of human donors	4 (ages 23–29)	4 (ages 49–54)	The biological replicate unit
Number of tissue samples	4	4	One per donor
Nuclei profiled, snRNA-seq		42,568 total	Nested within 8 donors
Nuclei profiled, snATAC-seq		41,550 total	Nested within 8 donors
Independent validation cohort		None reported	No second donor cohort for RNA/ATAC findings
External public datasets used	8 human tissues (cross-tissue); GTEx; oocyte scRNA-seq (GSE155179)		Comparative, not replication, datasets
Direct experimental validation	RNAscope (n=4/group); stem-cell allele-specific expression (HELB only)		Two forms of validation, narrow scope

05 Single-Nucleus Multi-Omics Analysis

With the replication caveats on the table, I want to walk through what the multi-omics analysis itself found, because there is real signal here worth taking seriously.

Cell-type composition changes

Comparing the proportion of each cell type between young and aged ovaries in the snRNA-seq data, the authors report a significant decrease in granulosa and theca cells with age — consistent with the well-established decline in follicle number — along with decreases in blood and lymphatic endothelial cells, and an increase in epithelial cell proportion, which they interpret as a possible signature of cumulative ovulation-induced rupture and repair. Reassuringly, snATAC-seq data show almost identical compositional shifts, which is a genuine internal consistency check between the two assays even though they were performed separately. I do think it is worth flagging a subtlety the paper does not dwell on: a decrease in a cell type's *recovered proportion* in a nuclei suspension can reflect a real decline in that cell type's absolute abundance in the tissue, but it can also be shaped by differences in nuclei recovery, cell fragility, or the specific tissue regions sampled. Granulosa and theca cell loss is well-supported independently by decades of histological literature on follicle depletion, which makes me more confident in that particular composition shift; but I would still want spatial or histological quantification as a direct cross-check on the proportions reported from dissociated nuclei alone.

Coordinated transcriptomic and chromatin changes

The study identified 3,455 aging-associated DEGs across the eight cell types, with granulosa cells contributing the most. Of the "common" DEGs shared across at least four cell types, 218 genes were up and 182 were down in aged ovaries, including recognizable aging-pathway genes such as *RICTOR*, *IGF1R*, *MAP3K5*, and *APOE*. Gene Ontology analysis of these common DEGs pointed to nutrient sensing, cellular senescence, proteostasis, cellular communication, and mitochondrial function — categories that map cleanly onto the "hallmarks of aging" framework the paper explicitly draws from.

Cross-tissue comparison

To ask whether this coordinated aging signature is unusual to the ovary, the authors compared pairwise correlations of aging-related expression fold-changes across cell types within the ovary against the same calculation in eight other human tissues (adipose, blood, bone marrow, brain, breast, liver, lung, esophagus), using publicly available single-cell datasets matched as closely as possible in age range (20–30 versus 45–55 years). Ovarian cells showed moderate-to-strong pairwise correlation; the comparison tissues showed weak or negligible correlation. This is a striking result, and it is the empirical basis for the paper's language about the ovary aging in an unusually coordinated way across its cell types. I discuss the interpretive limits of this specific comparison — different studies, different donors, different technical pipelines — in Section 6, because it is central to how the paper frames mTOR as "ovary-specific."

Pathway prevalence: amplitude and prevalence ratio

To move from "many genes change" to "which pathways matter most for tissue-level aging," the authors scored 21 aging-hallmark pathways per cell using Seurat's module-scoring approach, then assigned each pathway two summary metrics: an *amplitude index* (the mean absolute change in pathway activity across cell types) and a *tissue prevalence ratio* (the proportion of cell types showing a significant change). Pathways with a prevalence ratio of at least 0.7 were called "tissue aging prevalent," yielding ten such pathways within the ovary. This is a reasonable way to summarize a lot of per-cell-type results into a tractable shortlist, but it is worth remembering that the 0.7 threshold is a choice, not a natural constant

— a pathway just below that cutoff is not necessarily biologically less important, and I would be curious how sensitive the "top 10" list is to nearby threshold values, which the paper does not report.

06 The "Ovary-Specific mTOR" Claim

The single line from the abstract most likely to be quoted out of context is that mTOR signaling is "a prominent ovary-specific aging pathway." I want to take this claim apart carefully, because I think it is simultaneously one of the more interesting findings in the paper and one of the easiest to overstate.

What was actually measured: gene-expression-based module scores for the mTOR signaling pathway increased with age across most ovarian cell types (Fig. 2e), and this held up under a moderated t-test with Benjamini-Hochberg correction. When the authors applied their amplitude/prevalence-ratio framework across the ovary and eight comparison tissues, mTOR signaling landed with a high prevalence ratio and a comparatively high amplitude specifically in the ovary — of the 21 hallmark pathways evaluated this way, oxidative phosphorylation was the most prevalent pathway across tissues generally, but mTOR stood out as unusually prevalent in the ovary relative to the other eight tissues examined. Separately, the authors excluded genes identified as age-associated in female non-ovarian tissues using GTEx version 8 from their ovarian aging-associated DEG list. The resulting 2,301 genes were described as ovarian aging-specific and were enriched for cytoplasmic translation, a process regulated by mTOR. This GTEx-based filtering step is a distinct analysis from the eight-tissue single-cell correlation comparison described above, drawing on a different, bulk-tissue reference dataset. Separately, RICTOR — a core mTOR complex 2 component — was validated at the RNA level using RNAscope in situ hybridization, showing significantly higher expression in aged ovarian tissue (n = 4 donors/group).

What this evidence supports, in my reading: a consistent, RNAscope-corroborated increase in mTOR-pathway-associated gene expression with age in the ovary, and a comparative finding that this signature is less apparent, or absent, in the eight other tissue datasets examined under this pipeline. That is a real and interesting result. What it does not, by itself, establish: mTOR-pathway *gene expression* increasing is not the same claim as mTOR *kinase activity* increasing — mTOR is heavily regulated post-translationally, through phosphorylation events that an RNA- or accessibility-based module score cannot see. None of the analyses here measure phospho-protein levels, direct kinase activity, or a functional consequence of altered mTOR signaling in ovarian cells. The paper is appropriately careful about this in its Discussion, noting that "further mechanistic studies with larger sample sizes, protein-level validations and functional assays are required to confirm causality," but that caveat lives in a single paragraph near the end, while "ovary-specific mTOR aging pathway" is the phrase in the abstract that a reader — or a downstream citation — is more likely to carry forward.

I also want to flag the specific evidentiary weight of the cross-tissue comparison itself, since "ovary-specific" is a comparative claim and therefore only as strong as the comparison. The eight non-ovarian datasets came from different original studies, likely with different sequencing depths, cell-type granularities, batch structures, and donor compositions than the ovarian dataset generated for this paper. A pathway failing to show the same coordinated signature in, say, the adipose dataset could reflect a real biological difference, or it could partly reflect that the adipose study had less statistical power, different age-matching, or different single-cell chemistry. The authors' own language — describing mTOR as emerging as an "ovary-specific aging-prevalent pathway" within their defined analytical framework — is more careful than a blanket claim that mTOR involvement is absent from all other human tissues during aging, and I think that distinction is worth preserving whenever this finding gets cited elsewhere.

WHAT WOULD STRENGTHEN THIS CLAIM

Phospho-proteomic or Western blot evidence of mTORC1/2 activity in ovarian tissue across age; functional perturbation (for example, rapamycin or INK128 treatment) in a human ovarian organoid or explant model rather

than only the mouse literature cited as background; and a re-analysis of the cross-tissue comparison using uniformly processed datasets to remove differences in study design as a confound.

07 CEBPD and Regulatory-Network Interpretation

The second headline mechanistic claim concerns CEBPD, a CCAAT/enhancer-binding protein family transcription factor. Using chromVAR motif-activity scores computed from the snATAC-seq data, the authors report that C/EBP family motif activity increased significantly with age in most cell types (all except immune and epithelial cells), and that CEBPD itself was highly expressed across cell types, with its expression changes tracking motif-activity changes with age specifically in granulosa, immune, and theca cells. They then built cell-type-specific transcription-factor regulatory networks by combining three pieces of evidence: chromatin co-accessibility (via Cicero-derived cis co-accessibility networks, or CCANs), overlap with ENCODE-annotated ovarian tissue enhancers and promoters, and presence of a TF's motif within those accessible regulatory regions. Genes meeting all these criteria near a given TF's accessible motifs were called that TF's "candidate targets." CEBPD emerged as a hub-like regulator in several of these per-cell-type networks (Fig. 4e–k), with predicted target genes enriched for mTOR signaling, MAPK signaling, and cellular senescence pathways.

I think this is a well-constructed computational pipeline, and the convergence between CEBPD's own expression change and its motif-activity change is a nice piece of internal consistency. But I want to be precise about what each layer of evidence actually shows, because it is easy to let the network diagrams in Figure 4 imply more direct mechanism than was measured. Motif accessibility means a DNA sequence resembling the CEBPD binding motif sits in an open chromatin region — it is not a direct measurement of CEBPD protein occupying that site. Co-accessibility (CCAN) captures statistical correlation between accessibility at different genomic regions across pseudo-bulked groups of cells, which is a reasonable proxy for regulatory linkage but is still inferred rather than observed physical looping. And "candidate target" status in this network is a computational label built from accessibility, motif presence, and expression correlation — not the product of a chromatin immunoprecipitation experiment, a CRISPR interference perturbation, or a reporter assay demonstrating that CEBPD actually drives those genes' expression during ovarian aging.

None of this means the CEBPD hypothesis is wrong — the multi-layered agreement between expression and motif activity is a genuinely reasonable basis for prioritizing it as a candidate regulator, which is exactly the kind of hypothesis-generation this atlas is well-suited for. But "CEBPD is an important regulator of aging-associated gene expression in the human ovary," as the Figure 4 title puts it, is currently a well-supported computational hypothesis, not an experimentally demonstrated regulatory mechanism. Validating it would most directly require CEBPD perturbation (knockdown, knockout, or CRISPRi) in an appropriate ovarian cell model, ideally paired with CUT&RUN or ChIP-seq to confirm direct binding at the predicted target loci — none of which this paper performs for CEBPD specifically (the experimental validation effort in this paper was instead directed at the HELB locus, discussed in Section 8).

08 ANM Genetics and the Meaning of "Causal Variant"

This section of the paper is, in my view, both the most methodologically layered and the most important to read stage-by-stage, because the word "causal" gets used loosely in a lot of GWAS-adjacent writing, and this paper is largely — though not entirely — careful about it.

The evidentiary chain

Starting from 290 previously reported ANM-associated GWAS lead variants (from Ruth et al., 2021), the authors identified co-inherited proxy variants using linkage disequilibrium (LD, $R^2 \geq 0.8$) in 1000 Genomes European-ancestry

genotypes, yielding 5,555 ANM-associated variants. Overlapping these with cell-type-specific putative enhancers and promoters (built from the CCAN/ENCODE overlay described in Section 7) produced 101 candidate functional variants belonging to 62 lead loci, many shared across multiple cell types. A parallel analysis using TOPMed as the LD reference gave a broadly consistent, larger set (9,687 variants; 231 candidate functional variants from 90 lead loci). Cell-type enrichment of ANM-associated variants was independently supported by two methods: MAGMA applied to the snRNA-seq gene-expression specificity data, and stratified LD-score regression (LDSC) applied to the snATAC-seq chromatin-accessibility data — both found ANM-associated variants enriched across almost all ovarian cell types, in contrast to ovarian-epithelial-cancer-associated variants (enriched specifically in epithelial cells) and PCOS-associated variants (enriched specifically in granulosa cells).

The *HELB* variant: the strongest single result in the paper

Focusing on variants shared across at least four cell types, six sat in DNA-damage-response (DDR) gene loci. The best-developed example is rs3741605, whose T allele is associated with delayed ANM and sits in the putative promoter of *HELB*, a gene encoding DNA helicase B, which negatively regulates homologous recombination. A gapped k-mer SVM model predicted that the delayed-ANM T allele would reduce transcription-factor binding at this site, and GTEx eQTL data independently showed that the T allele correlates with decreased *HELB* expression in human ovarian tissue — two computational lines of evidence pointing the same direction.

Critically, the authors did not stop at computational prediction. Using human embryonic stem cells (H9) and induced pluripotent stem cells (IMR90-derived) that were heterozygous for rs3741605 and a tightly linked coding variant (rs4430553, $R^2 = 0.99$ with rs3741605 in European populations), they differentiated these lines into four ovarian cell types (endothelial, smooth muscle, stromal, granulosa) and used allele-specific digital PCR to directly measure transcription from each allele. Across all four differentiated cell types, and independently confirmed in primary cumulus granulosa cells from four IVF donors, the T-linked allele produced significantly less mRNA than the C-linked allele. This is a genuinely strong piece of functional evidence — an experimentally measured, allele-specific transcriptional effect, not just a predicted one — and it is, appropriately, the one place in the paper where I would use the word "demonstrated" rather than "suggested" or "nominated."

Even here, though, I want to hold the claim to its actual scope. The allele-specific expression experiments demonstrated lower *HELB* transcript abundance from the haplotype carrying rs3741605-T, tracked through the tightly linked coding variant rs4430553. This provides strong functional support for a cis-regulatory effect at the *HELB* locus, but it does not isolate rs3741605 from other linked variants as the sole causal nucleotide — the digital PCR readout depends on rs4430553 precisely because it is a coding variant that can be distinguished at the mRNA level, and rs4430553 sits on the same haplotype as rs3741605 rather than being manipulated independently of it. What was not demonstrated is that this transcriptional effect is the mechanism by which this variant delays age at natural menopause in a living person — that final causal step, from "this haplotype lowers a DNA repair gene's expression in a dish" to "this is why some women reach menopause later," remains an inference supported by a plausible biological story (less *HELB*, better-preserved homologous recombination capacity, slower follicle depletion) rather than a directly observed in vivo consequence. The authors state this limitation themselves in the Discussion, noting that establishing causality for ANM "requires further investigation," including variant knock-in approaches and consideration of other variants in strong LD.

The *DEPTOR* variant: a more provisional example

A second variant, rs13263296, sits in a TSS-proximal enhancer of *DEPTOR* — a direct inhibitor of mTOR signaling, which nicely dovetails with the paper's broader mTOR narrative — and its T allele is associated with earlier ANM. The gkm-SVM model predicted this allele would enhance TF binding, and *DEPTOR* expression was indeed significantly

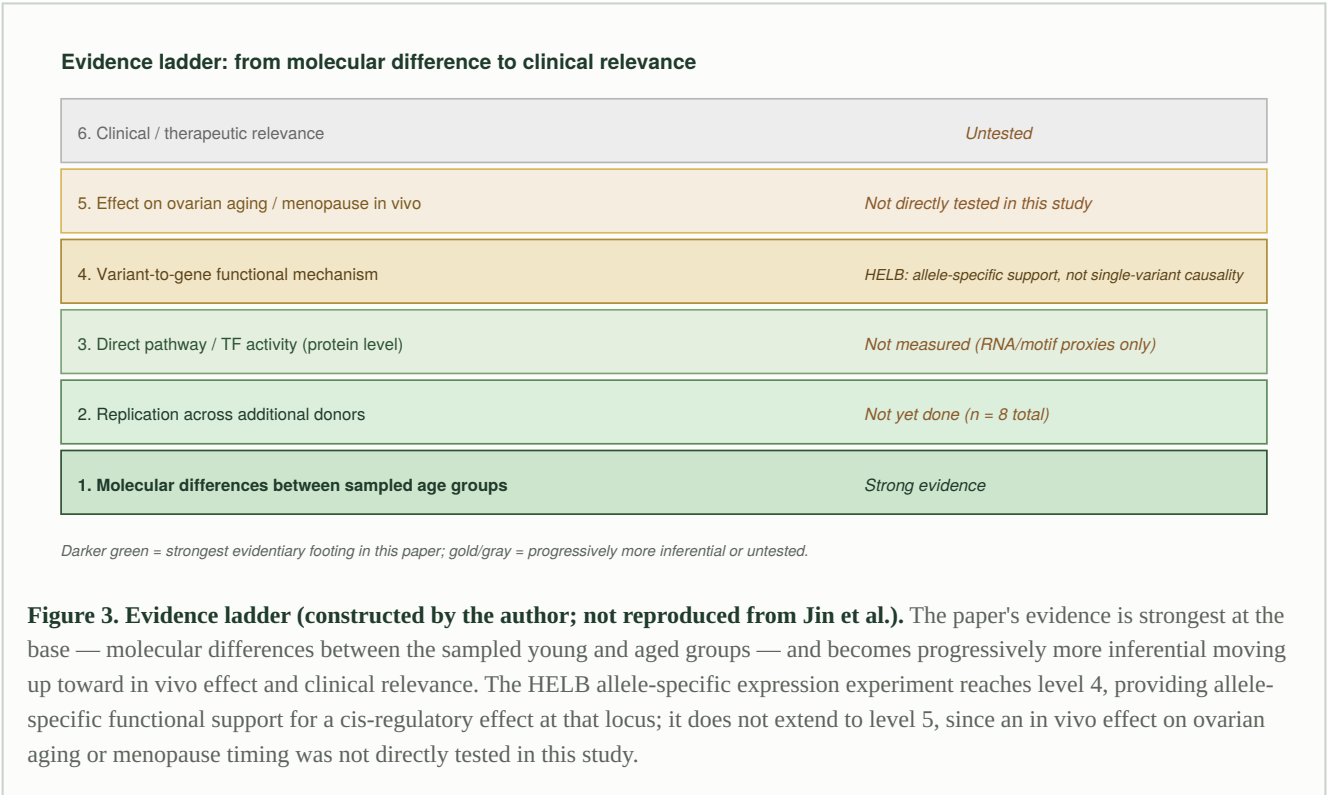
upregulated in aged granulosa and epithelial cells. However, the GTEx eQTL analysis for this variant in bulk ovarian tissue was not statistically significant ($P = 0.611$), which the authors attribute to the limits of bulk-tissue eQTL analysis for a cell-type-specific effect — a plausible explanation, but one that also means this candidate lacks the independent eQTL corroboration that strengthened the HELB story, and it received no stem-cell functional validation at all. I would treat DEPTOR as a well-motivated hypothesis rather than a confirmed mechanism, clearly one notch below HELB in evidentiary weight.

Ancestry and generalizability

The LD-based variant prioritization relied on European-ancestry reference panels (1000 Genomes and, in the alternative analysis, TOPMed), which is a common but real limitation for GWAS-adjacent work: LD structure differs across ancestries, so proxy variants and effect estimates derived from European-ancestry panels may not transfer cleanly to other populations. This is separate from, and should not be confused with, the ancestry of the eight ovarian tissue donors themselves, which is not reported in the sections of the paper I reviewed; I have not been able to verify donor ancestry and do not want to assume it.

Multiple testing: a specific, narrow caveat

One precise statistical point worth noting: the figure legends for the MAGMA and LD-score-regression cell-type enrichment analyses (Fig. 5a, 5b) report significance at $P < 0.05$ with the explicit note "no adjustment for multiple comparisons." This is a real limitation specific to those two enrichment analyses, tested across multiple cell types and multiple traits (ANM, PCOS, OEC) simultaneously — a setting where uncorrected testing does inflate false-positive risk. It would be inaccurate, though, to extend this criticism to the whole paper: the DEG, DAR, and pathway-activity analyses elsewhere in the paper explicitly use Benjamini-Hochberg or Bonferroni correction, as described in their respective Methods subsections.



09 Cell-Cell Communication and External Oocyte Data

To study how ovarian cells signal to one another and to the oocyte, the authors used CellChat, a tool that infers potential ligand–receptor communication from co-expression of known ligand-receptor gene pairs across cell types. The authors report that oocytes were not captured in their own datasets because of technical limitations, so the authors merged their own somatic-cell snRNA-seq counts with a separately published human oocyte single-cell RNA-seq dataset (GSE155179, from young and aged donors) as the oocyte input for CellChat.

The results are interesting and, I think, appropriately hedged in the paper's own language: total interaction number and strength decreased modestly with age; granulosa and theca cells received fewer signals from other cell types with age, as did endothelial and immune cells; signals reaching the oocyte from granulosa and theca cells specifically declined with age; and epithelial cells, in contrast, received more signaling input with age. At the pathway level, the authors identified 46 signaling pathways with significantly different communication probability between young and aged ovaries, including a decline in FSH and GDF signaling to granulosa cells and oocytes with age, paired with decreased expression of the FSH receptor (*FSHR*) and steroidogenic enzymes (*CYP11A1*, *CYP19A1*) in aged granulosa cells.

Two caveats are worth being explicit about. First, CellChat output is an inferred communication *probability* based on expression of ligand and receptor genes in the same dataset — it is not a direct, physical measurement of a signaling event occurring between two specific cells, and the paper's own figures correctly label this as "communication probability" rather than "observed signaling." Second, combining the study's own somatic-cell data with an externally published oocyte dataset introduces a layer of cross-study heterogeneity: the oocyte data came from a different research group, potentially differing in tissue handling, sequencing protocol, donor composition, age-group definitions, and batch structure relative to the ovarian somatic-cell data generated here. That does not make the oocyte-related findings wrong, but it does mean any oocyte-involving result in this analysis carries an extra layer of between-study uncertainty that the purely within-study somatic-cell comparisons do not. The paper is transparent that the oocyte data is external, which I appreciated when I was reading the Methods closely — it would have been easy to blur that distinction in the main text.

Table 1. Evidence and interpretation across the paper's major claims

Reported result	What it supports	What it does not establish	Best next test
3,455 age-associated DEGs across cell types	Coordinated transcriptomic differences between sampled age groups	Population-wide generalizability; timing/trajectory of onset	Larger, donor-diverse replication cohort
Chromatin accessibility changes (DARs)	Coordinated epigenomic differences paralleling expression changes	Causal direction between accessibility and expression change	Time-resolved or perturbation-based chromatin assay
mTOR pathway module scores ↑ with age; RICTOR RNAscope ↑	Increased mTOR-pathway gene expression in aged ovary	Increased mTOR kinase/protein activity; causal role in aging	Phospho-protein assays; functional perturbation in ovarian models
CEBPD motif activity & expression network	CEBPD as a strong computational candidate regulator	Direct CEBPD binding; experimentally confirmed target genes	CUT&RUN/ChIP-seq; CEBPD knockdown in ovarian cell model
ANM GWAS enrichment (MAGMA, LDSC)	ANM-associated genetic signals are enriched in ovarian cell-type gene-expression and chromatin-accessibility annotations	Individual variant causality; correction for multiple comparisons in this step	Multiple-testing-corrected replication; independent GWAS cohort
101 candidate functional ANM variants	Plausible, prioritized regulatory candidates	Proof of causal effect for most listed variants	Reporter assays / functional testing per variant
HELB allele-specific expression (stem cells + primary cells)	Lower allele-specific HELB transcript abundance from the haplotype carrying rs3741605-T in the tested cell models	That this mechanism changes ANM in vivo; that rs3741605 alone (versus other linked variants) is the causal nucleotide	Variant knock-in animal or organoid model; larger cohort replication
CellChat + external oocyte data	Plausible shifts in inferred cell-cell communication with age	Directly observed cell-to-cell signaling	Ligand-receptor functional or imaging-based validation

10 Strengths and Limitations

It would be easy, after four sections of close critique, to lose sight of how much this paper actually accomplishes. I want to lay out the balance sheet plainly.

Genuine strengths

- Access to rare, well-characterized human ovarian tissue with confirmed normal histology — a resource that is genuinely difficult to assemble at any donor number.
- Matched snRNA-seq and snATAC-seq generated from the same eight donor tissues, enabling a real (if not single-nucleus-paired) integration of transcriptomic and chromatin evidence.
- Broad, well-validated cell-type coverage across eight major ovarian somatic cell types, with detailed quality-control filtering reported transparently.

- A genuinely novel comparison against eight other human tissues, situating ovarian aging within a broader cross-tissue context.
- Integration with the largest available ANM GWAS, using two independent statistical frameworks (MAGMA and LDSC) that reached converging conclusions about broad cell-type enrichment.
- RNAscope validation of selected expression findings (RICTOR, MT-ATP6, CDKN1A) at the tissue level, beyond the sequencing data alone.
- A rare and well-executed piece of functional validation — the stem-cell allele-specific expression experiment at the HELB locus — going beyond computational prediction for at least one candidate variant.
- Public deposition of both raw/processed data (GEO GSE202601) and analysis code, supporting computational reproducibility.
- Overall, a valuable, hypothesis-generating atlas that did not previously exist for this tissue.

Real limitations

- Only eight biologically independent donors underlie every age-comparison claim in the paper, regardless of how many nuclei were profiled.
- A cross-sectional, two-group design cannot resolve trajectory, onset timing, or perimenopausal dynamics.
- Age, reproductive status, and follicle depletion are entangled in the "aged" group definition.
- Several cell/nucleus-level statistical tests do not clearly state donor-aware modeling, raising an unresolved (not confirmed) pseudoreplication concern.
- The central "ovary-specific mTOR" claim rests partly on gene-expression proxies for pathway activity and a cross-study, non-uniformly-processed tissue comparison.
- Most prioritized regulatory variants and CEBPD target genes remain computational hypotheses, not experimentally confirmed mechanisms.
- The MAGMA/LDSC cell-type enrichment analyses were not corrected for multiple comparisons.

Table 2. Strengths, limitations, and follow-up by study feature

Study feature	Strength	Limitation	Most useful follow-up
Donor cohort	Rare, well-characterized tissue; normal histology confirmed	n = 4 per age group; cross-sectional	Multi-site replication cohort, broader age range
snRNA/snATAC design	Matched donor tissue; large nucleus counts per cell type	Not true paired single-nucleus multi-omics	True co-assay (e.g., 10x Multiome) in future work
mTOR pathway finding	RNAscope-validated, cross-tissue comparative signal	No protein/phospho-level activity measurement	Phospho-proteomics; organoid perturbation
CEBPD network	Multi-layer computational convergence (motif + expression)	No direct binding or perturbation evidence	ChIP/CUT&RUN; CEBPD knockdown
ANM variant prioritization	Two independent enrichment methods; large GWAS input	Uncorrected P-values in enrichment step; Euro-ancestry LD	Multiple-testing correction; diverse-ancestry LD panels
HELB functional validation	Direct allele-specific expression in stem cells + primary cells	Not shown to affect ANM in vivo	Variant knock-in model; cohort replication
Data/code availability	Full public deposition (GEO, GitHub)	Reproducible pipeline ≠ independent biological replication	Independent cohort reanalysis using shared code

11 Reproducibility and Data Availability

The statistics and reproducibility statement is refreshingly candid: no statistical method was used to pre-determine sample size, no randomization method was used, and data collection and analysis were not performed blinded to condition — all of which the authors state directly rather than omitting. Normality of data distribution was assumed but not formally tested. Low-quality nuclei were excluded from both snRNA-seq and snATAC-seq analyses using pre-defined, clearly reported quality-control thresholds (described in Section 3).

On data availability: raw and processed snRNA-seq and snATAC-seq data are deposited in the Gene Expression Omnibus under accession GSE202601, and the external human oocyte single-cell RNA-seq data used for CellChat analysis are available under GSE155179. Analysis code is publicly available on GitHub, covering the snRNA-seq and snATAC-seq processing pipeline.

I want to draw a sharp line here that I think is easy to blur when reading a "strengths" list quickly: open data and open code are genuinely valuable and should be credited as such. Public data and code enable other researchers to attempt reproduction of the computational analyses and build on the processed datasets, rather than starting from scratch. Exact reproduction would still depend on the completeness of the released code, software environments, parameter settings, dependencies, and random seeds; I have not independently tested this. Separately, and more fundamentally, computational reproducibility of a pipeline is a different property from biological replication in an independent cohort: even a perfectly reproducible pipeline run on the authors' own published data would not tell you whether donor nine, sampled from a different population or clinic, would show the same granulosa cell mTOR signature. Both properties matter, and this paper's data and code release is a real strength on the first, while remaining, by its own design, unable to speak to the second.

12 My Critical Perspective

What initially impressed me about this paper was the sheer scope of the integration — RNA, chromatin, GWAS, cross-tissue comparison, and a wet-lab validation experiment, all organized around one coherent biological question. It is the kind of paper that makes we-can-connect-everything computational biology look genuinely productive rather than just impressive-looking. The part I found most difficult to interpret, on a first read, was exactly how much weight the "n = 8" was supposed to carry relative to the "n = 42,568" — the figures and the paper's framing elsewhere foreground the large nucleus counts, even though the abstract itself clearly states the donor numbers, and it took a careful pass through the Methods and statistics section to reconstruct where the donor-level replication actually sits in each specific analysis.

Reading the Methods changed how I understood the mTOR claim in particular. On a first read of the abstract, "ovary-specific aging pathway" sounded like a settled biological fact about mTOR and the ovary. After working through how the amplitude/prevalence framework was built, and what the cross-tissue comparison datasets actually were, I came away thinking of it instead as a well-supported, comparatively-framed hypothesis: mTOR-related gene expression rises with age in this ovarian cohort more consistently than in eight other tissue datasets processed by different groups under different conditions. That is still a genuinely useful finding — it is exactly the kind of signal that should motivate a targeted follow-up study — but it is a different-strength claim than "mTOR drives ovarian aging," and I think the paper's own framing occasionally lets the two blur into each other, particularly in the abstract and in some of the Discussion's more confident sentences about "potential interventions" and "geroprotectors targeting... mTOR signaling."

The HELB result, by contrast, is where my confidence in the paper's substantive claims is highest, precisely because the authors did not stop at prediction. Taking a computationally prioritized variant all the way to an allele-specific expression measurement in two independent stem-cell lines and four primary donor samples is a real commitment to functional validation that most GWAS-annotation papers do not make, and it changed how I weigh the rest of the paper's variant-prioritization work: I trust the pipeline that produced HELB more than I would have trusted it on computational prediction alone, even though the other 100 candidate variants have not received the same treatment.

My overall critical perspective, then, is not that this paper overclaims broadly — I actually think its Discussion is more careful than many papers I have read this term — but that the gap between donor-level replication (n = 8) and cellular measurement depth (tens of thousands of nuclei) is large enough that it deserves to be kept clearly in view throughout the paper's framing, not only in the Methods. Although the abstract clearly states the donor numbers, the large nucleus counts reported elsewhere should not be confused with the number of independent biological replicates.

13 What Would Change My Mind

I would raise my confidence in the paper's central mechanistic claims (ovary-specific mTOR involvement, CEBPD as a driving regulator, broad causal roles for the prioritized ANM variants) given any combination of the following:

- A replication cohort of additional donors — even a modest expansion to, say, twelve or sixteen per age group, ideally recruited from a different clinical source than BioIVT/Cureline — showing the same granulosa-cell composition shift and the same core DEG/DAR signatures.
- A donor-aware or pseudobulk reanalysis of the DEG, DAR, and differential motif-activity tests, explicitly modeling donor identity, that reproduces the paper's current significant findings.
- Protein-level or phospho-protein evidence of altered mTOR complex activity in aged ovarian tissue, not only RNA-based pathway scores.
- Direct evidence of CEBPD binding at its predicted target loci (ChIP-seq or CUT&RUN) in an ovarian cell context, or a CEBPD perturbation experiment showing predicted downstream effects.

- Extension of the cross-tissue mTOR comparison to a uniformly reprocessed set of tissue datasets, removing between-study technical variation as a possible confound.
- Functional or in vivo evidence connecting the rs13263296/DEPTOR mechanism to menopause timing, comparable in rigor to what was done for HELB.

Conversely, my confidence in the paper's more modest, descriptive claims — that ovarian cell-type composition and expression shift substantially between these particular young and aged donor groups, and that these shifts are internally consistent between the RNA and chromatin assays — is already fairly high, and would not need much more evidence to stay there; those claims are well within what an eight-donor, two-assay design can support.

14 Potential Research Extensions

- **Donor-scale expansion.** The single highest-value extension is simply more donors, ideally spanning intermediate ages (30s and 40s) rather than only the two sampled windows, to begin resolving trajectory rather than a two-point contrast.
- **Pseudobulk / mixed-effects reanalysis.** Applying donor-aware statistical models to the existing, publicly available processed data would directly test whether the current DEG/DAR/motif findings hold up when donor identity is explicitly modeled — this could be done without collecting any new tissue.
- **Spatial transcriptomics or histological quantification.** Directly validating the reported shifts in granulosa, theca, and endothelial cell proportions against spatial or immunostaining-based tissue quantification would separate true compositional change from dissociation/recovery artifacts.
- **Protein-level mTOR pathway assays.** Phospho-specific immunostaining or Western blotting for mTORC1/2 pathway components in the same or an expanded ovarian tissue cohort.
- **CEBPD functional perturbation.** CRISPR interference or knockdown of CEBPD in a differentiated stem-cell-derived ovarian cell model, paired with RNA-seq, to test predicted target-gene effects directly.
- **Diverse-ancestry variant prioritization.** Repeating the LD-based variant prioritization using non-European reference panels to assess how ancestry-dependent the current 101-variant candidate list is.
- **DEPTOR functional follow-up.** Applying the same stem-cell allele-specific expression strategy used for HELB to the DEPTOR-linked variant, to bring it to a comparable evidentiary standard.

15 Questions the Paper Leaves Open

- How much of the reported cell-type composition shift reflects true biological change in the intact ovary versus differential nuclei recovery during tissue dissociation?
- Would the DEG, DAR, and differential motif-activity results survive an explicit donor-level or pseudobulk statistical reanalysis?
- How sensitive is the "ovary-specific" mTOR finding to using uniformly reprocessed cross-tissue comparison data rather than eight separately generated public datasets?
- Does CEBPD actually bind its computationally predicted target loci in ovarian cells, and does perturbing it reproduce the predicted downstream expression changes?
- Among the 101 (or 231, under the TOPMed analysis) candidate functional ANM variants, which beyond HELB would survive direct functional testing?

- Would the DEPTOR/rs13263296 relationship hold up with a cell-type-resolved eQTL analysis, given that the bulk-tissue GTEx eQTL for this variant was not significant?
- How would the paper's cell-type-specific ANM enrichment results change under multiple-testing correction in the MAGMA/LDSC analyses?
- What does ovarian aging look like in donors of non-European ancestry, given that both the tissue procurement details and the LD-based variant prioritization lean on data and panels not obviously representative of that diversity?

16 Lessons Learned

The clearest lesson this paper taught me is that "sample size" is not a single number in single-cell and single-nucleus genomics — it is at least two numbers, and they answer different questions. The nucleus count tells you how well-resolved your cell-type measurements are; the donor count tells you how confidently you can generalize beyond the specific people you sampled. A paper can be simultaneously very large by one measure and very small by the other, and it is easy for a reader to let the more prominently repeated number — wherever in the paper it appears most often — carry more evidentiary weight than it should.

A second lesson was about how to read a "pipeline" of computational inference honestly. Nearly every mechanistic claim in this paper — mTOR's role, CEBPD's regulatory function, most of the 101 candidate ANM variants — is the product of several inferential steps stacked on top of each other (expression change, then pathway score, then cross-tissue comparison; or accessibility, then motif match, then co-accessibility, then network centrality). Each individual step is defensible, but the combined claim at the end of the chain is only as strong as its weakest link, and it is easy to lose track of how many inferential steps separate a headline claim from a direct measurement once the steps are presented as a finished network diagram.

A third, smaller lesson: reading the figure legends as carefully as the main text is where a lot of the real methodological detail lives — the "no adjustment for multiple comparisons" note on the MAGMA/LDSC figures, and the "n = 4 biologically independent samples" note on the RNAscope figures, are both details I would have missed if I had only read the Results prose.

17 Connection to My Development as a Bioinformatician

This is the fourth critical analysis in this portfolio, and I notice my reading habits have shifted a bit each time. Where I used to read a Results section mostly for "what did they find," I am now reading it more for "what is the unit of replication in this specific test," which this paper rewarded directly — its statistics and reproducibility statement is unusually explicit, and cross-checking that statement against each individual figure legend's stated n was, I think, the single most useful exercise in this analysis.

I would want to test my own understanding of this donor-versus-nuclei distinction by trying to build a small toy example myself: simulating a dataset with a handful of "donors" and many "cells" per donor, introducing a donor-level batch effect, and seeing how differently a cell-level test versus a pseudobulk or mixed-effects test behaves. I have not done that yet, but this paper is exactly the kind of real-world case that makes we want to.

I also came away with a more concrete appreciation for what "functional validation" can mean in practice, beyond the phrase itself. The HELB allele-specific expression experiment gave me a clear template — stem cell models, allele-specific PCR, primary cell confirmation — for what separates a computationally prioritized hypothesis from an experimentally supported one, which is a distinction I want to get better at applying critically in every paper I read from here on, not only this one.

18 Conclusion

Jin et al. have built a genuinely valuable, technically ambitious single-nuclei multi-omics atlas of the human ovary, generated from tissue that is difficult to obtain and paired with a comparatively rare piece of direct functional validation at the HELB locus. Their analyses generate well-supported hypotheses involving mTOR signaling, CEBPD regulatory activity, and ANM-associated regulatory variants — hypotheses that are worth taking seriously and worth funding follow-up work on. At the same time, the paper's cellular resolution, built from tens of thousands of nuclei, is substantially greater than its donor-level replication, which rests throughout on eight independent people. Consequently, I think this atlas is strongest exactly where the authors position it in their more careful moments: as a discovery and hypothesis-prioritization resource. Claims of ovarian specificity, population-wide aging mechanisms, and single-variant causality — including, at the level of pinpointing rs3741605 itself rather than the haplotype it sits on, the HELB result — would benefit from broader donor replication and further direct functional testing before they are treated as established biology rather than well-motivated leads.

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